機械力学 I 解答用紙

week-05 非減衰自由振動 - ばね質量系の運動方程式

学籍番号 20T1126N 氏名 YANG GUANGZE

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$X = \sqrt{x_0^2 + (\frac{v_0}{w_n})^2} = 0.326 [m]$

(1)	~~$X = \sqrt{x_0^2 + (\frac{v_0}{w_n})^2} = 0.326 [m]$~~
(2)	$\phi = \tan^{-1}(\frac{v_0}{x_0 w_n}) = -22.9 [rad]$

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(1)	$w_n = \sqrt{\frac{k}{m}} = 100 [rad/s]$
(2)	$f_n = \frac{w_n}{2\pi} = 15.9 [Hz]$
(3)	$T_n = \frac{1}{f_n} = 6.28 \times 10^{-2} [s]$

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(1)	$m\ddot{x} + (k_1 + k_2)x = 0$
(2)	$-m w_n^2 + k_1 + k_2 = 0$
(3)	$w_n = \sqrt{\frac{k_1 + k_2}{m}}$

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(1)	$k_e = \frac{2k(k+k)}{2k+k+k} = k$
(2)	$m\ddot{x} + kx = 0$
(3)	$-m w_n^2 + k = 0$
(4)	$w_n = \sqrt{\frac{k}{m}}$

以下余白